

Cybersecurity in Engineering

From digital threats to physical consequences

SAFETY

Can we operate safely?

INTEGRITY

Can we trust the system?

CONTINUITY

Can we keep operating?

What happens when equipment works, but its readings or controls cannot be trusted?

The route from cyber risk to engineering action

Three questions to carry through the presentation

UNDERSTAND

What are we protecting,
and from whom?

CONNECT

How do digital
dependencies affect
physical operations?

ACT

What can engineers
change in design,
maintenance, and
recovery?

You do not need to become a security specialist to make better security decisions.

What is cybersecurity?

Protecting information, systems, and the operations they enable

PEOPLE

Recognize unusual requests.
Assign owners and responsibilities.

PROCESSES

Approve changes.
Manage access.
Practice recovery.

TECHNOLOGY

Restrict connections.
Monitor activity.
Protect configurations.

Cybersecurity is a continuing risk-management activity.

When cyber incidents stop physical production

Two documented examples; different businesses, shared operational consequences

FAIRLIFE • JULY 2026

Ransomware-related unauthorized access included production-related systems.
U.S. production was temporarily suspended.
Company reported no product quality or safety impact.

JLR • SEPT-OCT 2025

Systems were proactively shut down after a cyber incident.
Retail and production were severely disrupted.
Phased restart included supplier cash-flow support.

A production stoppage can occur without confirmed damage to controllers.

Who creates the threat?

Motivation helps explain behavior

CYBERCRIMINALS

Money: extortion, fraud, or theft.

NATION-STATE ACTORS

Espionage, strategic access, or disruption.

HACKTIVISTS

Political or ideological objectives.

MALICIOUS INSIDERS

Misuse of legitimate access or knowledge.

Common entry methods: phishing • stolen credentials • exposed services • compromised suppliers

Five terms that explain cyber risk

Illustrative scenario: vendor access to an engineering workstation

ASSET	The workstation and the functions it supports.
THREAT	An attacker uses a stolen vendor account.
VULNERABILITY	Access is excessive or poorly controlled.
RISK	Likelihood and consequences of unauthorized changes.
CONTROL	Restricted, approved access and monitored sessions.

An incident is an event that compromises—or threatens—security; a breach involves unauthorized access or disclosure.

The CIA triad: what must be preserved?

Apply each principle to the same engineering system

CONFIDENTIALITY

Protect restricted information.

Example: credentials and network drawings.

INTEGRITY

Preserve accurate, authorized information and logic.

Example: relay settings and PLC programs.

AVAILABILITY

Keep required functions accessible when needed.

Example: operator visibility and control.

Safety can depend on all three. Priorities depend on the process and its hazards.

IT and OT: connected, with different constraints

OT monitors or controls physical processes

INFORMATION TECHNOLOGY

Examples: email, identity, business applications.

Common concerns: data protection and business continuity.

Changes still require risk assessment and testing.

OPERATIONAL TECHNOLOGY

Examples: PLCs, HMIs, SCADA, digital relays.

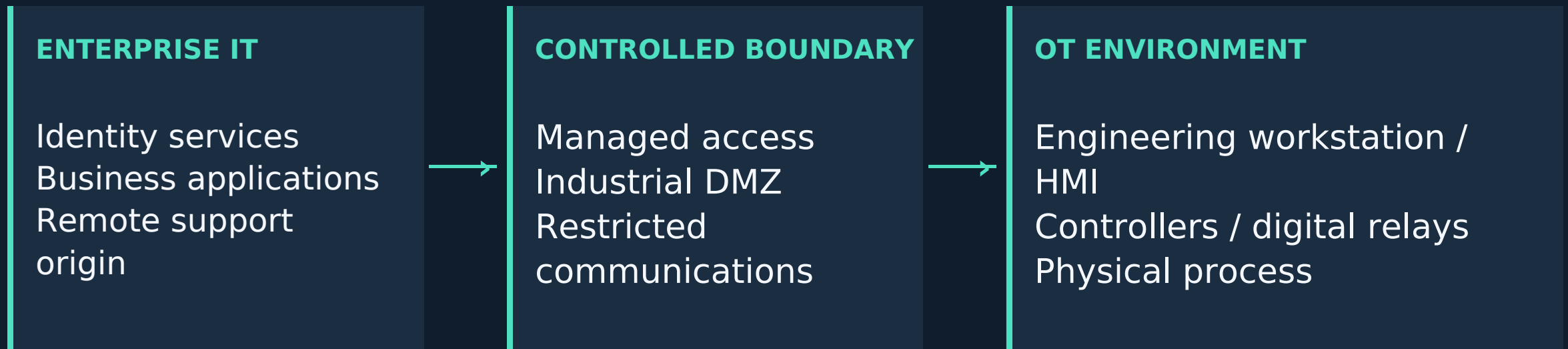
Physical consequences and timing can constrain changes.

Long service lives and limited outage windows are common.

HMI = human-machine interface • PLC = programmable logic controller • SCADA = supervisory control and data acquisition

How IT and OT interact

Conceptual architecture—not a site design or an attack reconstruction



Selected data paths: historian replication • approved vendor access • operational dependencies

Control the connections and understand the dependencies—not just the devices.

Defense in depth—and a plan to recover

Reduce entry opportunities, limit spread, and restore safely

KNOW THE ENVIRONMENT

Inventory assets, owners, connections, and dependencies.

CONTROL ACCESS

Verify identity; restrict privileges and remote access.

LIMIT & OBSERVE

Segment networks; collect useful logs and investigate anomalies.

RECOVER THE PROCESS

Preserve configurations; test restoration and safe restart.

A backup file is useful only if you can restore a trustworthy, functioning system.

Frameworks and standards: where they fit

Use one organizing framework; choose supporting guidance for the environment

NIST CSF 2.0

Organizes cybersecurity risk outcomes across an organization.

NIST SP 800-82

Guidance for securing operational technology.

ISA/IEC 62443

Standards for industrial automation and control system security.

ISO/IEC 27001

Requirements for an information security management system.

Govern

Identify

Protect

Detect

Respond

Recover

NERC CIP adds applicable electric-reliability obligations; it is not a universal OT framework.

Greenfield and brownfield sites

The same security goals; different opportunities and constraints

GREENFIELD • NEW DEVELOPMENT

Specify security during procurement and design.

Plan segmentation and managed vendor access.

Test security and recovery during commissioning.

BROWNFIELD • EXISTING INSTALLATION

Validate existing assets and connections.
Phase improvements around outage windows.

Manage interfaces between old and new equipment.

Brownfield does not automatically mean legacy; greenfield does not automatically mean secure.

Legacy environments: managing the constraints

Age is a clue; support, capability, and consequence determine the risk

CHALLENGES

- Unsupported software.
- Limited security features.
- Difficult replacement.

ENGINEERING RESPONSE

- Restrict access and communications.
- Monitor critical paths.
- Preserve recovery capability.

CHANGE CONSTRAINTS

- Test compatibility.
- Coordinate outages.
- Confirm vendor support.

LONG-TERM PLAN

- Assign an owner.
- Prioritize by consequence.
- Fund staged replacement.

Compensating controls reduce risk; they do not make unsupported technology equivalent to supported equipment.

AI: useful assistance, new trust decisions

Treat generated output as a proposal that needs validation

DEFENDER USES

Summarize logs.
Draft documentation.
Assist analysis and review.

ADVERSARY USES

Create convincing impersonation.
Scale deceptive messages.
Assist malicious workflows.

ENGINEERING GUARDRAILS

Use approved tools and data.
Verify code and calculations.
Keep humans accountable.

Do not connect generated code or autonomous actions to live equipment without approved review and testing.

NERC CIP: cybersecurity for electric reliability

Critical Infrastructure Protection within a defined regulatory scope

PURPOSE

Protect reliable operation of the Bulk Electric System (BES).

APPLICABILITY

Determine covered entities, facilities, and systems under the standards.

CATEGORIZATION

CIP-002 identifies and categorizes BES Cyber Systems by impact.

High, medium, and low impact are formal categories—not general labels for every electrical asset.

Where engineers contribute to NERC CIP

Useful engineering records support security and compliance

ASSET KNOWLEDGE Maintain accurate equipment, ownership, and connectivity records.

ACCESS & VENDORS Coordinate physical access and approved remote connections.

CHANGE & EVIDENCE Preserve settings, approvals, drawings, and implementation records.

INCIDENTS & RECOVERY Know contacts; preserve evidence; support safe restoration.

Low-impact CIP-003-9 plans cover awareness, access, incident response, transient devices/media, and vendor remote access.

One decision: a vendor wants remote access

Discussion scenario: troubleshooting a digital relay at an operating site

BEFORE

Verify the person and ticket.
Approve scope, duration, and operational conditions.

DURING

Use the approved access path.
Limit permissions.
Maintain required visibility.

AFTER

Revoke temporary access.
Verify settings and operation.
Retain relevant records.

Ask the audience: "What would you verify before approving this connection?"

Three things to take back to your project

Cybersecurity belongs in engineering decisions throughout the lifecycle

KNOW

Know the assets, owners, and dependencies.

CONTROL

Control access and changes.

PROVE

Verify recovery and trustworthy operation.

Questions? Think of one design or maintenance decision where these ideas apply.

Selected sources

Primary sources • checked October 6, 2026

FOUNDATIONS

- [1] NIST CSF 2.0
- [2] NIST SP 800-82 Rev. 3 (final)
- [9] ISA/IEC 62443 series

INCIDENTS

- [3] Coca-Cola / fairlife disclosure
- [4] JLR initial statement
- [5] JLR restart announcement

ELECTRIC RELIABILITY

- [6] NERC CIP directory
- [7] CIP-002-5.1a
- [8] CIP-003-9

Full links are in speaker notes and the presenter guide. Incident examples do not establish an attack path.